

Epic MegaGrants Official Application Form Draft

This document contains pre-filled responses formatted specifically for the official **Epic MegaGrants** online application portal fields (unrealengine.com/megagrants).

1. Project Information

Project Name

Riva — Deterministic Performance Diagnostics Companion for Unreal Engine

Primary Category

Unreal Engine Tools & Open Source Components

Developer Portfolio & Website

<https://darwinstylish.dev>

Repository URL

<https://github.com/DarwinStylish/Riva>

Primary Contact Email

darwin@darwinstylish.dev

License Type

MIT License (Open Source)

2. Short Project Description (100–150 Words)

Riva is an open-source, deterministic performance diagnostics companion for Unreal Engine. While Unreal Insights collects rich, multi-thread trace recordings, game developers often spend hours manually scrubbing timeline tracks to understand frame hitches.

Riva automates bottleneck isolation by processing .utrace recordings and JSON trace exports through a 6-stage C++20 diagnostic pipeline. It isolates frame spikes, evaluates built-in signatures (GC, Shader Compilation, PSO Misses, Streaming IO, RHI Waits, Lumen/VSM GPU variance), graph-links cascading thread dependencies to elevate root causes over symptoms, and clusters repetitive spikes.

Riva includes a native Unreal Editor Slate companion tab (SRivaPanel) with bi-directional Unreal Insights timeline selection sync, as well as a standalone headless CLI (riva) for automated performance budget gatekeeping and regression diffing in CI/CD build pipelines.

3. Detailed Technical Description & Architecture

Riva is designed around a strict decoupling model split into two primary components:

- **riva_core (Pure C++20 Static Library):**
 - Contains zero Unreal Engine header dependencies for maximum portability and instant compilation in headless build environments.
 - Converts raw trace inputs into a unified NormalizedTrace data model.
 - Runs diagnostic analysis through 6 sequential stages:

SpikeDetector $\rightarrow$ ISignature Rules $\rightarrow$ CausalChainResolver $\rightarrow$ ConfidenceResolver $\rightarrow$ CorrelationResolver $\rightarrow$ FReportEngine / ITraceComparator.

- **RivaEditor (Native Unreal Engine 5 Editor Plugin):**
 - Implements FRivaTraceService to ingest native .utrace recordings via Unreal Engine's TraceServices module.
 - Provides a dockable Slate panel (SRivaPanel) under Window -> Developer Tools -> Riva Performance Diagnostics.
 - Executes trace analysis asynchronously on Unreal's background thread pool (EAsyncExecution::ThreadPool) to keep the editor UI 60 FPS responsive.
 - Links time range selections bi-directionally with Unreal Insights tracks.

- **riva (Standalone CLI Executable):**
 - Implements `riva analyze`, `riva compare` (regression diffing), and `riva check-budget` (CI gatekeeper returning exit code 3 on metric breaches).
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4. Community & Developer Impact

Riva directly benefits the entire Unreal Engine ecosystem:

- **Indie Developers & Small Studios:** Eliminates the steep learning curve of manual timeline scrubbing, giving small teams AAA-grade performance diagnostic capabilities directly in the editor.
 - **Enterprise & AAA Game Studios:** Integrates into automated PR pipelines to prevent "performance decay" during multi-year development cycles.
 - **Open-Source Community:** Provides an extensible `ISignature` C++ framework for creating custom diagnostic rules (e.g. Niagara VFX, Chaos Physics, World Partitioning).
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5. Current State & Proof of Concept

Riva is **not an unproven concept**—it is a working, production-grade codebase:

- **Developer Website:** darwinstylis.dev
 - **Repository:** [GitHub - DarwinStylis/Riva](https://github.com/DarwinStylis/Riva)
 - **Test Suite:** 13 CMake / CTest test executables passing with a 100% success rate.
 - **Documentation:** Comprehensive documentation suite including User Guides, Architecture Specifications, and Mermaid diagrams.
 - **Sample Datasets:** Includes 8 sample trace datasets covering Shader Compiles, PSO Misses, Streaming IO, GC, RHI Sync, and Lumen GPU variance.
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6. Requested Funding Amount & High-Level Purpose

Requested Amount

\$50,000 USD

Primary Purpose

To fund full-time development over a 6-month roadmap focused on expanding native `.ut` trace event extraction (World Partitioning, Chaos Physics, Niagara), implementing automated call-stack symbolication (PDB/dSYM), building a standalone cross-platform GUI application, and packaging Riva as an officially distributed Marketplace & GitHub Action tool.